

the network. The first step is to connect the webcam and see whether Raspberry Pi recognises it—if you see a file called ‘video0’ under ‘/dev’, everything should work well!

Next, you can install the popular ‘ffmpeg’ package, which does a whole lot of video manipulations, including streaming, using the following command:

```
pacman -S ffmpeg
```

Let’s now create a small configuration file called ‘ffserv.conf’. This file supplies information like the network port at which the stream is available, the name of the stream, etc. Issue the following commands:

```
Port 8010
BindAddress 0.0.0.0
MaxClients 10
MaxBandwidth 1000

<Feed feed1.ffmpeg>
File /tmp/webcam.ffmpeg
FileMaxSize 5M
</Feed>
```

```
<Stream feed1.mjpeg>
Feed feed1.ffmpeg
Format mjpeg
VideoSize 160x128
VideoFrameRate 3
VideoIntraOnly
Noaudio
Strict -1
</Stream>
```

Check out <http://ffmpeg.org/sample.html> for more information on the parameters in this configuration file. Streaming of the video is done by a program called ‘ffserver’. Invoke it with the following command:

```
ffserver -f ./ffserv.conf
```

‘ffserver’ automatically goes into the background. Now you have to run ‘ffmpeg’ to capture data from the webcam and supply it to ‘ffserver’, as follows:

```
ffmpeg -v 2 -r 5 -s 160x128 -f video
o4linux2 -i /dev/video0 http://localhost:8010/feed1.ffmpeg
```

You can open the browser on a remote machine and enter the following URL to watch a streaming video: <http://192.168.1.5:8010/feed1.mjpeg>. (You have to replace the address 192.168.1.5 with the IP address assigned to Raspberry Pi’s Ethernet interface.)



Figure 2: Boot messages scrolling by on the screen

How to move your webcam with a servo motor

You can mount your webcam on the shaft of a hobby servo motor and control it using Raspberry Pi! The easiest way to do this is to run the servo control code on Arduino and send commands to Arduino from Raspberry Pi. When Arduino is connected to the USB port of Raspberry Pi, the serial port on Arduino is accessible as a special file (usually ‘/dev/ttyACM0’) on Raspberry Pi. The Python module ‘pyserial’ can be used to write the code, which sends and receives commands over this serial channel. (In our case, the ‘command’ may be simply a number that indicates the angle at which the servo shaft should be positioned.)

The servo control code, as well as the Python code running on Raspberry Pi, are available for download at: <https://github.com/rlabs/rpi-article>.

The start of a revolution!

The Raspberry Pi revolution has just begun. This tiny credit-card sized GNU/Linux system has generated tremendous enthusiasm among hobbyists and ‘do it yourself’ fans all over the world. You can already see people coming up with amazing ideas like the ‘fishpi’ (<http://fishpi.org/>)—an autonomous Raspberry Pi controlled marine vehicle that its designers plan to launch on a journey across the Atlantic, unaided! 

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